ATM Transaction Analytics with SQL, Python and Power BI

PhD Carlos Eduardo Romero Figueroa

Synthetic ATM dataset, January–June 2025 (103? no: 12,000 transactions across 30 ATMs)

Executive summary. We analyze ATM transaction behavior and operational performance using SQL Server for KPI extraction, Python for exploratory analysis and forecasting, and Power BI for decision-oriented visualization. The synthetic generator uses a realistic intraday demand profile with stronger morning and evening activity; this is a modeling assumption rather than evidence about a real ATM network.

1. Data and Business Objective

The dataset contains **12,000 transactions** across 30 ATMs. The overall rejection rate is **3.57%**, approved withdrawals total **HNL 8,346,300**, and mean transaction latency is **7.74 seconds**. The main questions are when demand is highest, where cash volume is concentrated, and which ATMs combine service-quality issues with meaningful usage.

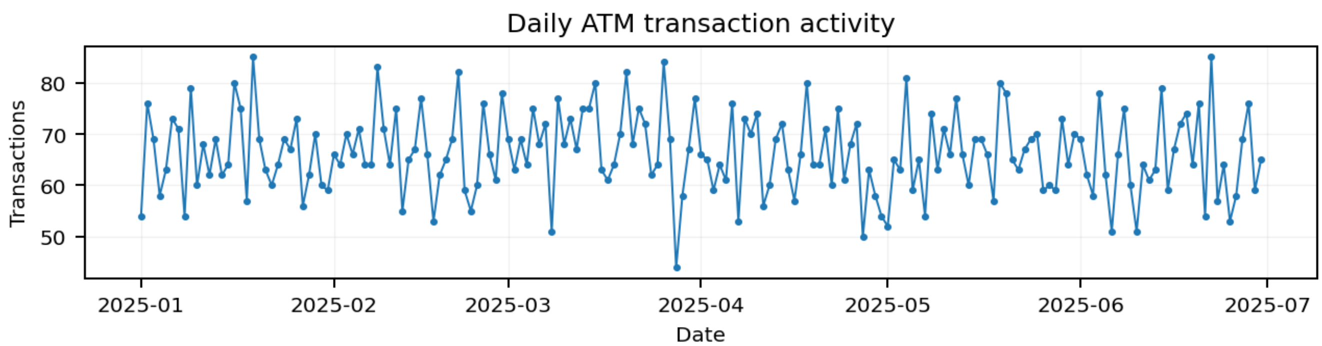

Figure 1. Daily ATM transaction activity.

2. Demand Analysis

Hourly demand is concentrated in business and commuting periods rather than overnight hours. In the simulated dataset, the busiest hour is **18:00**, with strong activity also around the morning and evening peaks. This profile is intentionally designed to resemble a plausible ATM usage pattern for a portfolio exercise; it should not be presented as empirical evidence.

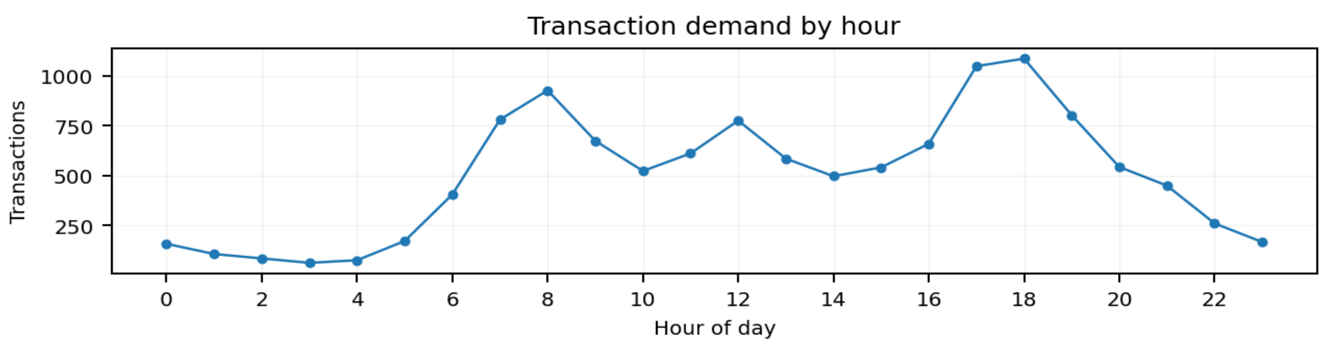

Figure 2. Transaction demand by hour of day.

3. Geographic and Cash-Volume Analysis

Tegucigalpa is the highest-volume city by approved withdrawal amount in this synthetic sample. City-level aggregation helps prioritize cash logistics because transaction counts alone do not measure the amount of cash moving through the network.

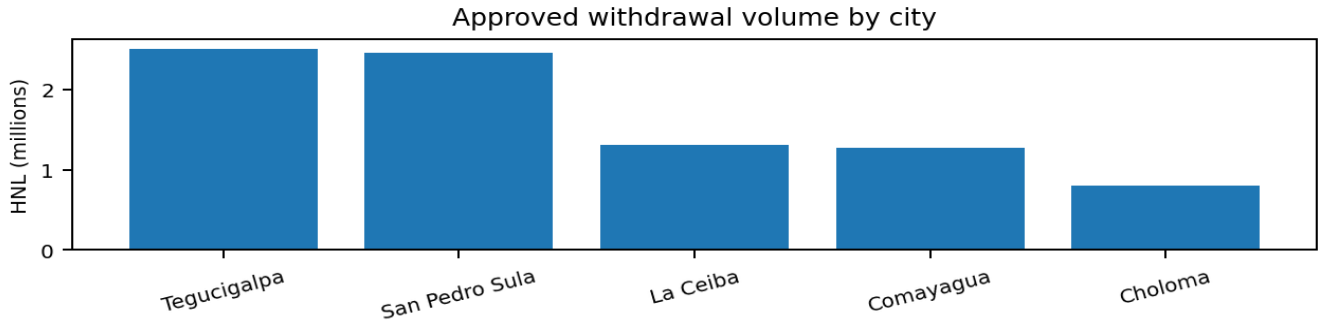

Figure 3. Approved withdrawal volume by city.

4. Operational Risk by ATM

ATM-level statistics expose differences hidden by network-wide averages. The highest observed rejection-rate ATM is **ATM-018**. A useful operational rule is to prioritize machines showing elevated rejection rates and latency, especially when transaction volume is also high.

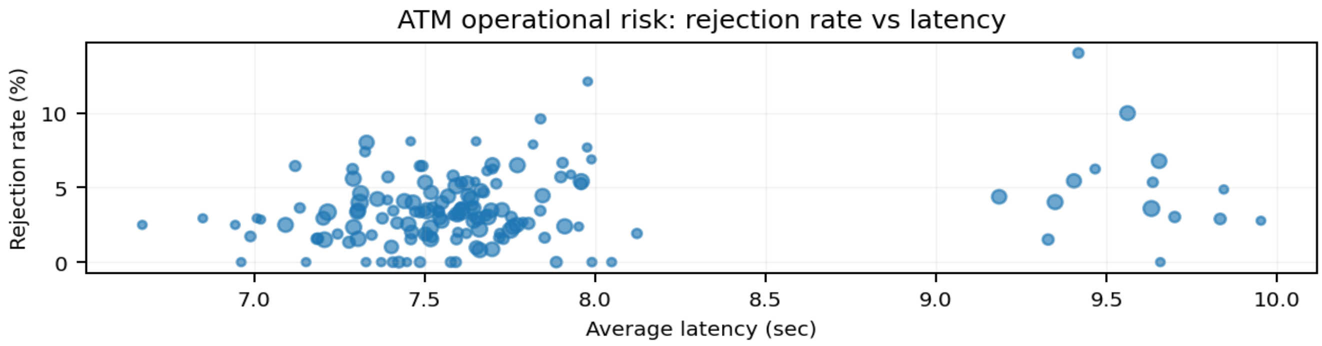

Figure 4. Rejection rate versus latency; point size represents transaction volume.

5. Predictive / Monitoring Layer

Daily approved withdrawal volume can feed a forecasting workflow for cash replenishment planning. Candidate production models could include exponential smoothing, SARIMA, Bayesian time-series models, or machine-learning models with calendar and operational features. Evaluation should use chronological holdout data and metrics such as MAE and RMSE.

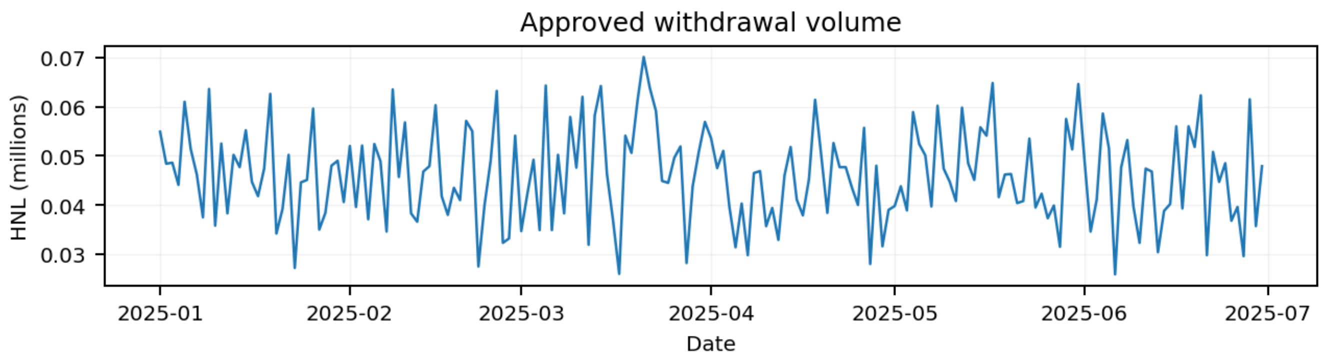

Figure 5. Approved withdrawal volume over time.

6. SQL + Python + Power BI Workflow

SQL Server: transactional storage, aggregations and operational KPIs. **Python:** data validation, exploratory analysis, risk scoring and forecasting. **Power BI:** executive dashboard with KPI cards, demand trends, city volume, ATM rankings, and rejection-versus-latency analysis.

7. Conclusions

The synthetic network demonstrates how raw ATM transactions can be transformed into operational indicators. Demand peaks at **18:00** in the simulated data, while city and ATM analysis reveal concentration in cash volume and service-quality differences. The forecasting layer provides a direct bridge from descriptive analytics to predictive cash-demand planning. All numerical results are synthetic and intended for portfolio demonstration.